

CASE REPORT



Nutritional ultrasound: above body mass index, two cases presentation

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Malnutrition is a prevalent condition in chronic diseases, significantly impacting morbidity and mortality. Point-of-care ultrasound (POCUS) is increasingly utilized in clinical practice as a rapid and accessible tool for evaluating muscle mass. Here, we present two cases of females with chronic diseases who presented with acute exacerbations of their conditions accompanied by fluid overload. Suspected of experiencing nutritional alterations, they underwent ultrasound evaluation to confirm muscle mass loss. These cases highlight the potential of ultrasound in guiding effective muscle mass assessment, particularly in pathologies prone to fluid overload, such as chronic kidney disease and heart failure.

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INTRODUCTION

Nutritional assessment using ultrasonography of the quadriceps rectus femoris (QRF) may have a pivotal role. Malnutrition, prevalent in chronic conditions exerts a profound impact on patient outcomes, contributing to heightened morbidity and mortality rates. Assessing muscle mass loss is a crucial alternative indicator considering the limitations associated with Body Mass Index (BMI) measurements, especially influenced by fluid overload. In clinical practice, the growing use of point-of-care ultrasound (POCUS) is transforming the way health professionals assess patients in various settings and across a wide range of clinical scenarios. Ultrasound emerges as a highly valuable and practical tool for assessing muscle mass, with broad applicability across diverse clinical scenarios. The highlighted cases underscore the urgent need for early malnutrition detection and the timely implementation of nutritional interventions.

CASE 1

A 65-year-old female presented to the emergency department with a two-day history of bilateral leg edema, increased peritoneal diameter, and intermittent dyspnea. Upon admission, she displayed palpebral edema, bilateral lung crackles, and bilateral leg edema with fovea. Further investigations revealed elevated creatinine (1.8 mg/dL), hypoalbuminemia (2.9 g/dL), dyslipidemia (total cholesterol: 261 mg/dL), and a 24-hour urine collection test reported a protein excretion of 3.17 g. Etiologic studies ruled out autoimmune and infection causes. A percutaneous biopsy of the kidney showed a diabetic nephropathy. Management for chronic kidney disease was initiated.

The patient presented a height of 156 cm, an actual weight of 62.8 kg, and a BMI of 24.1 kg/m². Her usual weight was reported as

60 kg, reflecting a weight gain of 4.6%. She reported a decrease in food intake of 30% in the last 2 weeks. The patient denied any decrease in food intake over the last week. Nutritional ultrasound showed a QRF area of 143 cm², confirming muscle mass lost (Fig. 1).

CASE 2

A 34-year-old female presented to the emergency department for bilateral leg edema and orthopnea. She has a history of hypertrophic cardiomyopathy and underwent heart transplantation in 2019. Despite four episodes of antibody-mediated rejections, and interventions including plasma exchange therapy and intravenous immunoglobulin, she experienced a decline in ejection fraction. Therefore, it was considered to have acute heart failure, and diuretic therapy with furosemide was initiated. However, in the following days she showed no improvement in congestive signs, with kidney function markers alteration (creatinine: 3.7 mg/dL) and hyponatremia suggesting hypervolemia. Peritoneal dialysis was started.

Her height was 159 cm with an actual weight of 56 kg for a body mass index of 21.48 kg/m². Her usual weight was 54 kg for a net weight gain of 3.6%. In the last week she decreased 20% of food intake. Nutritional ultrasound indicated a QRF area of 2.11 cm², confirming decreased muscle mass (Fig. 1).

Two cases are presented, from Bogotá, Colombia. They had chronic diseases with weight gain attributed to fluid overload. Nutritional Ultrasound was performed in a supine position, cross-sectional area of the QRF without compression with a 10 to 12 MHz probe from a multifrequency linear matrix. Cut-off points were extrapolated from other studies. This method is a key assessment to evaluate body composition and confirm muscle mass loss, and it must be added to adequate nutritional evaluation.

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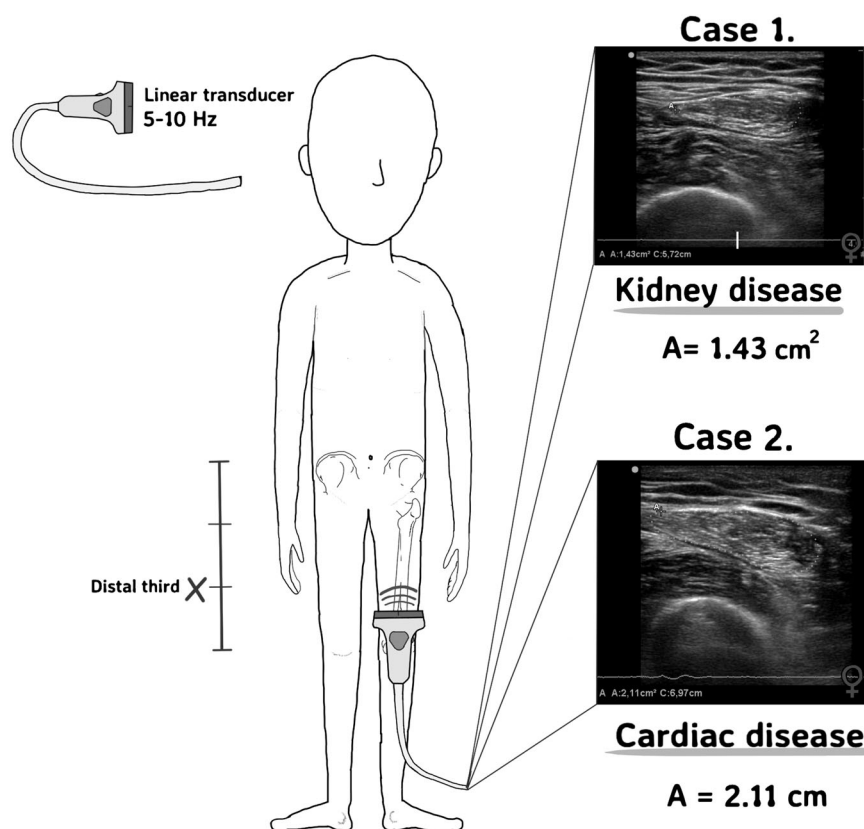


Fig. 1 Visual representation illustrating the nutritional ultrasound procedure, including anatomical localization, transducer specifications, and image findings from the cases. Image created using Tayasui Sketches.

DISCUSSION

The prevalence of hospital malnutrition substantially increases the susceptibility to infectious and non-infectious complications, leading to higher mortality rates and extended hospitalization periods [1]. This elevated risk is associated with a surge in healthcare costs, ranging from 38% to 77% [2, 3]. Malnutrition is closely intertwined with chronic diseases, including heart failure and chronic kidney disease, as the patients noted [4]. Apart from BMI and the percentage of weight loss, muscle mass loss is also included in the phenotypic criteria of malnutrition [4]. Notably, BMI measurements can be influenced by fluid overload, making it sometimes unreliable for malnutrition diagnosis. This underscores the importance of considering muscle mass loss as an alternative indicator. Nutritional ultrasound for QRF protocol and technique has been described [5]. Thus, it appears as an accessible, fast, and practical tool that allows for a reliable estimation of nutritional status.

The AnyVida Trial in 57 cancer patients, the non-survivor patients had an area of 2.98 cm²/m² in contrast with 4.27 cm²/m² of the survivors ($p = 0.03$), and those with a high muscle area had 49% less mortality (HR: 0.61, 95% CI: 0.39–0.96, $p = 0.031$) [6]. The VALIDUM study in the intensive care unit, although they used the thickness of the rectus femoris area, found a good relationship compared with the abdominal muscle cross-sectional area measurement by computer tomography, which may be more precise in young males ($p < 0.001$) [7]. Among hemodialysis patients, those with a low cross-sectional area had 8 times higher risk-adjusted (OR: 8.63, 95% CI: 4.80–15.50, $p < 0.001$) of developing protein-energy wasting in hemodialysis [8].

Despite the increasing number of publications of this assessment modality, there still missing cut-offs, especially in Latin America. In 118 internal medicine patients, the cut-off

points were for men $< 7.2 \text{ cm}^2$ (AUC: 0.755) and for women $< 4 \text{ cm}^2$ (AUC: 0.937) [9]. Among 351 hemodialysis patients a cut-off of less than 6 cm² or in women of 4.47 cm² had a sensitivity of 62% and 77% and a specificity of 80% and 80% respectively [8]. In cancer the AnyVida Study established a cut-off point of $\leq 4.47 \text{ cm}^2$ [6]. The 2 patients had an area of less than 2.2 cm² which is less than 45% of the described cut-off and was extrapolated.

Even more, a systematic review that included 32 ultrasound-guided skeletal muscle assessment found major protocol variability [10]. Some authors propose to unify and standardize this body composition evaluation [5]. It is still pending the results for the Spanish multicenter prospective trial (NCT05433831): The DRECO study, which will evaluate the muscle ultrasound evaluation with diagnosis and malnutrition follow-up [11].

CONCLUSION

The presented cases underscore the prevalence of muscle mass loss in chronic diseases and emphasize their relevance to clinical physicians. Muscle ultrasonography emerges as a crucial tool, providing a practical and effective method for assessing muscle mass loss, a key indicator of malnutrition. These cases lend support to the GLIM criteria's emphasis on muscle mass evaluation, revealing the inadequacy of relying solely on BMI. The findings underscore the necessity of standardizing nutritional assessment methods, particularly in diverse clinical scenarios within an upper-middle-income economy. A validation study in Colombia is essential to establish specific cutoff points. Ultrasound, with its advantageous features such as portability and accessibility across various clinical scenarios, holds promise as a simple yet potent and advanced tool for exploring body composition.

DATA AVAILABILITY

The project adheres to Resolution 8430 of October 4, 1993, of Colombia and had the verbal and written consent of the patient for publication as well as approval from the Fundación Cardio Infantil Ethics Committee.

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AUTHOR CONTRIBUTIONS

All authors contributed to the study conception and design. Data collection and analysis were performed by S.G.J., S.S. and, J.C. The first draft of the manuscript was written by S.G.J. and S.S. Review and editing was made by E. Z., J.C., A.M. and, S.G.J. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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COMPETING INTERESTS

The authors declare no competing interests.

ETHICS STATEMENT

The authors declare that the procedures were followed according to the regulations established by the Clinical Research and Ethics Committee and to the Helsinki Declaration of the World Medical Association updated in 2013.

CONSENT FOR PUBLICATION

Informed consent was obtained from all patients.

ADDITIONAL INFORMATION

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